

TEB 2023/ TFB 2073/ FEM 3013: ARTIFICIAL INTELLIGENCE

SEPTEMBER 2026 SEMESTER

ASSIGNMENT LAB 07

Tutor Name: Ms Bag Fiza

No	Student Name	Matric ID	Course
1.	Shanchanaa Subramaniam	24006193	Bachelor of Computer Science

Part A - Understand the concepts

Q1. NLP pipeline

Describe the five stages of an NLP pipeline: acquire, normalise, represent, model and evaluate. For each stage, give an example of what happens when a handwritten medicine-name image is processed.

Stage 1 (Acquire):
The raw data is collected. For a handwritten medicine name, the prescription is scanned or photographed and the image file is loaded into the computer
Stage 2 (Normalise):
The data is cleaned so that everything is consistent. For the image, this means converting it to grayscale, adding a border and resizing it. For the text, it means lowercasing the string and removing spaces and symbols. So “Tab Azimax 40mg” becomes “tabazimax40mg”
Stage 3 (Represent):
The data is turned into a form the computer can work with. The image becomes a grid of pixel numbers and the text becomes a string of characters
Stage 4 (Model):
The OCR engine (Tesseract) reads the image and produces text. For example, it reads a photo of a handwritten prescription line and outputs “tab azimax”
Stage 5 (Evaluate):
The output is compared with the correct reference text using metrics such as exact-match accuracy, Levenshtein distance and Character Error Rate (CER). For example, comparing the output “nidu” with the reference “nims” shows how many characters are wrong

Q2. How OCR works

Explain the difference between segmentation and character recognition. Give one example of a failure at each stage. Why is it useful to identify which stage caused an error?

Segmentation means finding and separating the parts of the image that contain text such as lines, words or single letters. Character recognition means looking at each separated part and deciding which character it is

A segmentation failure happens when two neighbouring letters are treated as one. For example, “cl” may be read as “d”. A character recognition failure happens when a letter is separated correctly but the handwriting looks like a different character. For example, a handwritten “s” may be read as the digit “3”. In the lab, “nims” being read as “nidu” is another example of this

It is useful to know which stage caused an error because it tells us what to fix. Segmentation problems need better image preparation and recognition problems need a better model or a dictionary of known words

Q3. Text normalisation

The reference is “Tab Azimax 40mg” and the OCR output is “tab azimax 40 mg”. Apply the Lab 6 normalisation rule (lowercase and keep only letters and digits). Show both normalised strings. Why must the same rule be applied to both?

The rule is applied in two steps. First lowercase everything, then remove everything that is not a letter or digit

Reference: Tab Azimax 40mg → tab azimax 40mg → tabazimax40mg
OCR output: tab azimax 40 mg → tabazimax40mg

The same rule must be used on both strings so that the comparison is fair. Otherwise, differences in capital letters or spaces would be counted as errors even though the text is the same

Q4. Normalisation trade-off
The Lab 6 normaliser removes spaces. Describe one situation where removing spaces could hide an important difference. Suggest how you would change the normalisation rule for that situation.
Removing spaces can hide the boundary between separate numbers. For example, “5 10 mg” and “510 mg” both become “510mg” eventhough they mean very different doses. This could hide a serious dosage error

To fix this, I would change the rule to keep spaces but make them consistent. Lowercase the text, keep only letters, digits and single spaces, collapse repeated spaces into one and remove spaces at the start and end. The downside is that “40 mg” and “40mg” would then count as a small difference

Part B — Work with OCR measurements

Q5. Levenshtein distance
Calculate the Levenshtein distance between “breeky” (reference) and “beeky” (prediction). Identify the edit operation(s). Then calculate the normalised edit distance by dividing the distance by the length of the longer string.
Reference: "breeky" (length = 6)
Prediction: "beeky" (length = 5)
Edit Operations: 1 Deletion (the character r is deleted from breeky to get beeky)
Levenshtein Distance: 1
Normalised Edit Distance = (Distance)/ (length of longer string)
= 1/max(6,5)
= 1/6
= 0.167

Q6. Character Error Rate (CER)
An OCR system produces 4 substitutions, 2 deletions and 1 insertion. The reference contains 35 characters. Calculate CER using CER = (S + D + I) / N. Explain what the result means. Can CER be greater than 1? Explain briefly.
Substitutions (S) = 4, Deletions (D) = 2, Insertions (I) = 1, Reference Length (N) = 35
Calculation:
CER = (S + D + I)/N
= (4+2+1) /35
= 7/35
= 0.200
This means the number of errors equals 20% of the reference length or roughly 1 error for every 5 characters in the reference

Yes, CER can be greater than 1. This happens when the total number of errors is bigger than the reference length. For example, if the reference has 5 characters and the OCR outputs 15 characters of random text, the errors can add up to more than 5 so CER is above 1

Q7. Compare the metrics
Explain exact-match accuracy, mean normalised edit distance, CER and blank predictions. For each metric, state whether a higher or lower value is generally better. Why is exact-match accuracy alone insufficient?

Metric	Better when	What it measures
Exact-match accuracy	Higher	The proportion of images where the normalised prediction is exactly the same as the normalised reference
Mean normalised edit distance	Lower	For each image, the edit distance is divided by the length of the longer string. These values are then averaged across all images
Character Error Rate (CER)	Lower	The total number of edits (substitutions + deletions + insertions) divided by the total number of characters in the references
Blank predictions	Lower	The number of images where the OCR returned an empty string

Exact-match accuracy alone is not enough because it only says whether the answer is completely right or completely wrong. A small mistake like “azimaxx” instead of “azimax” scores the same as a blank output or random symbols. It also cannot show partial improvement. In the lab, exact-match only moved from 0.000 to 0.011, but blank predictions dropped from 42 to 15 and CER fell from 0.883 to 0.806. Those other metrics show progress that exact-match missed

Part C — Interpret the Lab 6 experiment

Use the results from Lab 6. Both pipelines were evaluated on the same 89 labelled images, with the same normalisation rule and fixed Tesseract configuration.

Q8. Read the results
Describe at least two improvements shown by the preprocessed results. Use values from the table as evidence.
The preprocessed pipeline which includes grayscale conversion, border addition and resizing improved on the baseline in three measurable ways which are:

1. Blank predictions fell from 42 to 15. This is 27 fewer images with no output (42 − 15 = 27). This is a reduction of 64.3% (27 ÷ 42 = 0.643). It means Tesseract produced some output for many more images
2. Character Error Rate (CER) fell from 0.883 to 0.806. This is a reduction of 0.077 or 7.66 percentage points which is an improvement of about 8.7%
3. Mean normalised edit distance fell from 0.869 to 0.785. This is a reduction of 0.084

Lower values are better for all three metrics so each change indicates an improvement as seen in Fiugre 1

	pipeline	exact_match_accuracy	mean_normalised_edit_distance	character_error_rate	blank_predictions
2	baseline	0.0	0.8687316428889463	0.8831168831168831	42
3	grayscale_border_resize	0.011235955056179775	0.7850723870386791	0.8064935064935065	15

Figure 1: Baseline vs Preprocessed Results

Q9. Explain the result pattern
CER and mean normalised edit distance improved, and blank predictions decreased. However, exact-match accuracy increased only from 0.000 to 0.011. Explain how preprocessing can help recover text without making the complete medicine name correct.
Preprocessing helped Tesseract produce output for images where it previously produced none which is shown by the fall in blank predictions from 42 to 15. A blank prediction is the worst possible result for an image with a normalised edit distance of 1.0. When a blank becomes even a partly correct output, the average error falls which explains the lower CER and mean normalised edit distance

However, exact-match accuracy requires every character of the medicine name to be correct. Handwritten names are often cursive with connected letters and Tesseract is not designed for this style of writing. In the error analysis, “tab novidat” was read as “| i» welll” and “tab distalgesic” as “wr”. In these cases some output was produced, but it was still very different from the reference. The CER of 0.806 also shows that errors remained high overall. As a result, only 1 of 89 images (0.011) was read completely correctly eventhough the other metrics improved

Q10. Controlled experiment
Why did the lab keep the Tesseract settings (--oem 1 --psm 7), the 89 images and the normalisation rule unchanged? What would be difficult to conclude if several settings changed at once?
The lab kept the Tesseract settings (--oem 1 --psm 7) the same 89 images and the same normalisation rule unchanged so that the only difference between the two pipelines was the image preprocessing. This makes any change in the results attributable to preprocessing alone which is the basis of a fair comparison

If several settings had changed at the same time, it would be difficult to conclude which change caused the difference in performance. For example, a change in the normalisation rule alters the scores without changing the OCR quality at all and using different images would make one pipeline’s task easier or harder than the other’s. The results would then be confounded and the improvement could not be credited to any single change

Q11. Diagnose the failure
Identify the likely category for each case and explain:
(a) Tesseract raises an error while processing an image
Category: technical or processing error
This is not a reading mistake. The process fails before any text is produced. For example, because the image file is corrupted, missing or in an unsupported format. The cause is the file or the software not the handwriting

(b) OCR returns an empty string
Category: no text detected
Tesseract ran successfully but found no recognisable text in the image. This may be caused by unclear or faint handwriting, a poor-quality image or a failure to locate the text. The empty output alone does not show which of these was the cause so further inspection of the image would be needed

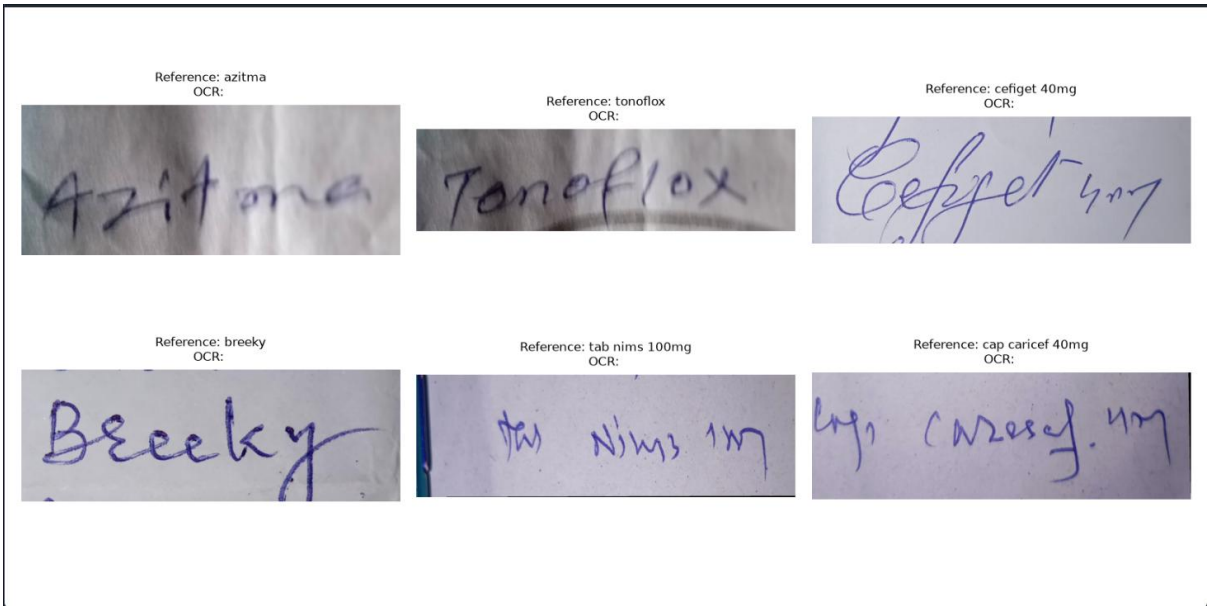


Figure 2: Six handwritten images from the Lab 6

(c) OCR returns "nidu" when the reference is "nims".
Category: character confusion
The text was found and the first two letters were correct but "m" was read as "d" and "s" was read as "u". This is two substitutions in a four-letter word giving a normalised edit distance of $2 \div 4 = 0.50$

Part D — Apply your understanding

Q12. Suggest improvements
Suggest TWO improvements you would investigate next to improve handwritten OCR. For each, explain which error it targets and how you would test whether it helps. Do not claim an improvement works without evidence.
1) Adaptive thresholding

- This method converts each image to pure black and white using a separate brightness threshold for each small region. Uneven lighting and shadows therefore have less effect on the result
- The error it targets are blank predictions on images with shadows, creases or low contrast
- How to test:
  Apply this step to the same 89 images and keep the Tesseract settings (--oem 1 --psm 7) and the normalisation rule unchanged. Compare the results with the current grayscale_border_resize pipeline: 15 blank predictions, CER 0.806, mean normalised edit distance 0.785 and exact-match accuracy 0.011. The method would be considered helpful only if these values improve. If they do not, the result should be reported as no improvement

2) Dictionary-based correction of the OCR output

- After OCR, each output is compared with a list of valid medicine names. If it is very close to one of them, it is replaced by that name
- The error it targets are character confusion where the output is close to the correct

name but has a few wrong letters. For example “diclorew” instead of “dicloran”

- How to test:
  Build the name list from a source that is independent of the 89 test labels. Otherwise the test would be unfair because the correct answers would be included in the list. Then measure exact-match accuracy, CER and mean normalised edit distance again and count how many corrections became correct and how many became wrong. Its benefit is not assumed until these results are measured
- Limitations:
  This method cannot fix blank outputs or unrelated symbols such as “wr”. A wrong replacement could also produce a different but valid medicine name which is a safety concern

Q13. Medical safety scenario

A clinic wants to use OCR to read medicine names and dosages from handwritten prescriptions. Explain at least THREE risks of using OCR output without human checking. Recommend a safer workflow.

1) Dosage errors
- A small misreading of a number or a decimal point such as “40 mg” read as “400 mg” or “4.0 mg” read as “40 mg” could cause a serious overdose. In the lab, the reference “tab azitma 500mg” was read as “thy Badan Seq” so the dosage was lost completely

2) Confusion between similar medicine names
- Names that look alike when handwritten can be read as one another which would lead to the wrong treatment. Character confusion errors such as “nims” read as “nidu” show how easily letters are misread

3) Missing information
- A blank prediction means an instruction or medicine could be left out without any warning. Even after preprocessing, 15 of the 89 images still produced blank output

Recommended safer workflow:
- Use OCR only as an assistant that suggests text not as a system that makes decisions by itself
- Mark results with low confidence and all blank results for mandatory review by a pharmacist
- Show the original handwritten image next to the OCR text so that a pharmacist or clinician can compare them and must confirm or correct each entry before it is saved

Q14. Reflection

What is the most important thing you learned from Lab 6? Identify one concept you initially found difficult and explain it using an example from the practical.

The most important thing I learnt is that a standard OCR engine performs poorly on handwriting and preprocessing improves the results only modestly. After preprocessing, CER was still 0.806 and only 1 of 89 images was read exactly. This means OCR output should not be trusted without checking,particularly for medical use. I also learned that several metrics are needed to understand how well a system performs

I initially found it hard to understand why CER and normalised edit distance can improve while exact-match accuracy stays close to zero.

Example:
- For the reference “bisleri”, the baseline gave a blank output which has a normalised

edit distance of 1.0. After preprocessing, the output was “Bus ld” and the distance fell to 0.571 (4 edits ÷ 7 characters). The output was still wrong but it was closer to the reference

- For “dicloran”, the baseline output was “divlerer” (distance 0.500). After preprocessing, the output was “diclorew” (distance 0.250). Two characters were still wrong so the exact-match result remained 0

Edit-based metrics reward this kind of partial progress while exact-match accuracy does not